

Conflict-Resilient Multi-Agent Reasoning via Signed Graph Modeling

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Abstract

LLM-based multi-agent systems (MAS) have demonstrated strong reasoning and decision-making capabilities that consistently surpass those of single LLM agents. However, their performance often suffers from naive aggregation mechanisms that assume uniformly cooperative interactions. Upon close inspection, we observe that existing graph-based MAS frameworks ① propagate errors when conflicting signals arise without control, and ② lack explicit modeling of conflicting inter-agent relations as well as structural awareness, failing to identify reliable interaction patterns. To bridge this gap, we introduce **SIGMA**, a novel *Signed Graph-informed Multi-Agent reasoning framework* that explicitly captures *trust*, *conflict*, and *neutral* relations among agents via a signed relational graph. Specifically, given a query, **SIGMA** first selects a set of relevant and diverse agents, then constructs a structured signed interaction graph with confidence-weighted edges. Reasoning proceeds through conflict-aware signed message passing, which reinforces information from trustworthy agents while suppressing conflicting signals, and terminates with a structure- and conflict-aware weighted aggregation to yield globally consistent and conflict-resilient predictions. Extensive experiments on six benchmark datasets, across multiple LLM backbones and diverse multi-agent configurations, demonstrate that **SIGMA** consistently outperforms state-of-the-art baselines, achieving notable gains in both accuracy and conflict-resilient performance.

1 Introduction

As Large Language Models (LLMs) continue to reshape the landscape of artificial intelligence, *LLM-driven agents* have demonstrated remarkable capabilities in reasoning [1, 2, 3, 4], planning [5, 6, 7], and decision making [1, 8, 9], exhibiting increasing autonomy and adaptability across a wide range of applications, including code generation [10, 11], data analysis [12], and embodied intelligence [13]. Building upon the success of single-agent systems, recent studies have shown that LLM-based Multi-Agent Systems (MAS) can further enhance performance by leveraging collaborative intelligence [14, 15, 16]. By orchestrating multiple agents with diverse expertise [17, 18], MAS enables more robust and scalable problem-solving, shifting from isolated reasoning to collaborative intelligence.

Existing MAS methods can be broadly categorized into three types based on the interaction topology among agents: *Chain-based* [19, 20, 21], *Tree-based* [22, 23, 24], and *Graph-based* [25, 26, 27, 28] approaches, as illustrated in Figure 1 (Left). *Chain-based MAS* organize agents sequentially, forming a simple reasoning pipeline, whereas *Tree-based MAS* employ hierarchical coordination to aggregate outputs from multiple branches. *Graph-based MAS* represent agents as nodes and interactions as edges, explicitly modeling inter-agent dependencies and enabling flexible information flow [4, 29]. Among these three types, *graph-based MAS* has attracted considerable attention due to its capability to capture complex interactions and support more adaptive multi-agent reasoning.

However, existing graph-based multi-agent system methods still exhibit notable limitations. First, they cannot explicitly model conflicting relationships between agents. Most methods rely solely on generic similarity or communication links to construct the graph [25, 26, 30], which leads to implicit conflict signals propagating uncontrollably across multiple reasoning steps, thereby amplifying errors. Second, they lack logical consistency and structural awareness, making it difficult to identify and leverage the latent patterns in agent interactions [27, 28], which in turn undermines system robustness and reliability in complex tasks. Together, these limitations leave existing graph-based MAS methods particularly vulnerable in scenarios involving noisy or adversarial agents.

To address this, we propose a *Signed Graph-informed Multi-Agent Reasoning framework* for LLM-based multi-agent systems, dubbed SIGMA. As illustrated in Figure 1 (Right), unlike prior graph-based MAS approaches, SIGMA captures not only the existence but also the polarity of agent interactions via signed graph modeling [31, 32, 33]. Edges encode not only the existence of interactions but also their polarity [34] (e.g., *trust* or *conflict*), allowing SIGMA to characterize multifaceted inter-agent relationships through three complementary types of relations: ① *Trust*, capturing supportive interactions; ② *Conflict*, modeling contradictory interactions; and ③ *Neutral*, representing weak or uncertain interactions. By explicitly modeling interaction polarity, SIGMA identifies trustworthy agents and those to challenge, enabling robust and consistent multi-agent reasoning.

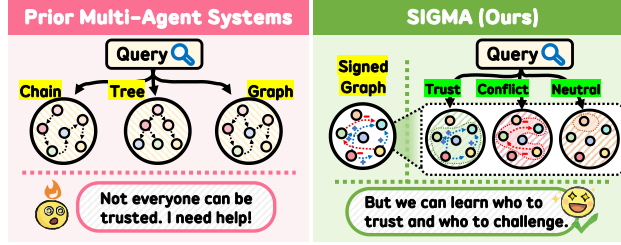


Figure 1: (Left) Prior MAS treat all agents as equally reliable, including chain, tree, and graph-based structures. (Right) SIGMA models trust, conflict, and neutral relations via signed graph modeling, enabling the system to identify which agents to trust or challenge for conflict-resilient reasoning.

Although the approach is promising, introducing signed graph modeling is non-trivial and involves three key modeling challenges. ① *More complex interactions*. Agent interactions are highly dynamic and multifaceted, generating supportive, conflicting, or neutral signals that propagate across multiple reasoning steps. If not properly handled, these interactions can amplify errors or suppress valuable dissenting information. ② *More vulnerability to noise or conflicts*. The presence of low-confidence or conflicting agent outputs increases the likelihood of misleading signals affecting the final consensus. Without careful management, noise can propagate through the network, reducing overall reliability. ③ *More challenging information aggregation*. Aggregating positive and negative signals across multi-hop neighborhoods is challenging. Improper handling may lead to information collapse, overlooked conflicts, or inconsistent global representations, ultimately compromising multi-agent reasoning.

To tackle these challenges, SIGMA first performs query-guided agent selection, ensuring that the chosen agents are both semantically relevant and diverse. It then constructs a signed heterogeneous relational graph, estimating pairwise agreement signals and annotating each relationship as trust, conflict, or neutral, with corresponding confidence weights. Reasoning proceeds via a conflict-aware signed message passing mechanism, which reinforces information from trusted agents while suppressing conflicting signals, thereby mitigating the influence of unreliable agents. Finally, a structure- and conflict-aware weighted aggregation integrates all agent outputs to maximize agreement, minimize inconsistency, and produce a globally consistent, high-quality prediction. By explicitly modeling *trust*, *conflict*, and *neutral*, SIGMA transforms inter-agent interactions from uniform cooperation into structured reasoning dynamics, enabling more reliable and robust multi-agent LLM reasoning.

Contributions. We summarize our main contributions as follows: ① We highlight a key limitation in existing multi-agent systems: assuming uniformly cooperative interactions without modeling trust or conflict, which reduces robustness under contradictory or noisy outputs. ② We propose SIGMA, a *signed graph-informed MAS framework* that models *trust*, *conflict*, and *neutral* relations, enabling conflict-resilient multi-agent reasoning. ③ Extensive experiments show that SIGMA consistently outperforms baselines, achieving robust predictions even with conflicting or low-quality agents.

Organization. The rest of this paper is organized as follows. In Section 2, we introduce the notations and preliminaries. Section 3 details our proposed method SIGMA. Section 4 presents comprehensive experimental results. In Section 5, we discuss related work. Finally, Section 6 concludes the paper.

2 Preliminary

In this section, we formalize *LLM-based multi-agent systems* and extend the interaction modeling with a *signed graph representation* to explicitly model both trustworthy and conflicting interactions.

LLM-based Multi-Agent System Formalization. Consider a multi-agent system of LLM-based agents [1, 2, 14] modeled as a directed interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $|\mathcal{V}| = N$ denotes the number of agents and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents communication links. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ denote the interaction matrix, where $\mathbf{A}_{ij}$ quantifies the strength of influence from agent v_j to agent v_i . Each agent $v_i \in \mathcal{V}$ is instantiated by a LLM with distinct roles, prompts, or reasoning strategies [15, 25]. We abstract each agent as a computational unit capturing its reasoning and interaction processes:

$$v_i = (\mathcal{M}_i, \mathcal{R}_i, \mathcal{P}_i), \quad (1)$$

where $\mathcal{M}_i$ denotes the underlying LLM, $\mathcal{R}_i$ specifies the role configuration, and $\mathcal{P}_i$ defines the prompting strategy. Given a query Q , the system evolves over T interaction rounds. At each iteration t , each agent maintains a latent reasoning state $\mathbf{h}_i^{(t)} \in \mathbb{R}^d$, which captures its intermediate reasoning output. These states are updated via interactions with neighbors. The update rule is defined as:

$$\mathbf{h}_i^{(t)} = f_i\left(Q, \mathbf{h}_i^{(t-1)}, \sum_{j \in \mathcal{N}(i)} \mathbf{A}_{ij} \mathbf{h}_j^{(t-1)}\right), \quad (2)$$

where $f_i(\cdot)$ denotes the reasoning function parameterized by the LLM, and $\mathcal{N}(i)$ denotes the neighborhood of agent v_i . Let $\mathbf{H}^{(t)} = [\mathbf{h}_1^{(t)}, \dots, \mathbf{h}_N^{(t)}]^\top \in \mathbb{R}^{N \times d}$ denote the global matrix. The system dynamics can be expressed in compact matrix form across all agents over interaction rounds as $\mathbf{H}^{(t)} = \mathcal{F}(Q, \mathbf{H}^{(t-1)}, \mathbf{A})$, where $\mathcal{F}(\cdot)$ aggregates agent-wise updates in a unified iterative message-passing process. After T rounds, a global aggregation operator $\mathcal{A}(\cdot)$ produces the final prediction: $y = \mathcal{A}(\mathbf{H}^{(T)})$. Most existing methods [25, 26, 27] assume a non-negative interaction matrix ($\mathbf{A}_{ij} \geq 0$), implicitly assuming all interactions are cooperative. This assumption neglects the presence of conflicting or unreliable signals among agents, which may lead to unbounded error accumulation, degraded reasoning robustness, and unstable consensus formation in complex reasoning scenarios.

Signed Graph Representation. Inspired by Balance Theory [35], signed graphs represent relationships by polarity and magnitude. Specifically, each interaction is given by $\mathbf{A}_{ij} = s_{ij} \cdot w_{ij}$, where $s_{ij} \in \{-1, 0, +1\}$ indicates polarity and $w_{ij} \geq 0$ its magnitude, as shown in Definition 1.

Definition 1 (Balance Theory [35]). *A triad of nodes v_i, v_j, v_k is balanced if the product of its edge signs $\prod_{(p,q) \in \{(i,j), (j,k), (k,i)\}} s_{pq} = +1$ is positive; otherwise, it is imbalanced. Balanced triads reflect consistent configurations. For instance, if v_i has a positive relationship with v_j and a negative relationship with v_k , the triad is imbalanced. To restore balance, v_i may either form a positive relationship with v_k or a negative relationship with v_j . Figure 2 shows all four triad types.*

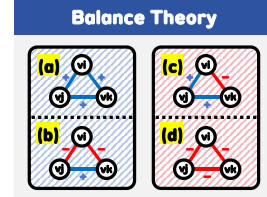


Figure 2: Visualization of Balanced Triad types. First two are balanced, last two are imbalanced.

Accordingly, the signed matrix admits a decomposition $\mathbf{A} = \mathbf{A}^+ - \mathbf{A}^-$, where $\mathbf{A}^\pm = \max\{\pm \mathbf{A}, 0\}$, with the max operator applied element-wise. This separates positive and negative interactions and induces a partition of each agent's neighborhood into $\mathcal{N}_i^\pm = \{j : \pm \mathbf{A}_{ij} > 0\}$. Under this signed structure, propagation distinguishes cooperative and conflicting signals [36]. A signed aggregation is:

$$\mathbf{h}_i^{(t)} = \sum_{j \in \mathcal{N}_i^+} \alpha_{ij} \mathbf{h}_j^{(t-1)} - \sum_{j \in \mathcal{N}_i^-} \beta_{ij} \mathbf{h}_j^{(t-1)}. \quad (3)$$

where α_{ij} and β_{ij} are normalized attention weights induced by $\mathbf{A}^+$ and $\mathbf{A}^-$, respectively. To ensure stable propagation, we further normalize the signed interaction matrix as $\tilde{\mathbf{A}}_{ij} = \mathbf{A}_{ij} / (\sum_k |\mathbf{A}_{ik}| + \epsilon)$, which controls the scale of incoming interactions and improves numerical stability across layers. This yields the following compact formulation for signed aggregation under normalized propagation:

$$\mathbf{H}^{(t)} = \tilde{\mathbf{A}}^+ \phi(\mathbf{H}^{(t-1)}) - \tilde{\mathbf{A}}^- \phi(\mathbf{H}^{(t-1)}). \quad (4)$$

where $\phi(\cdot)$ denotes a shared transformation over agent representations, implemented via LLM reasoning or prompt-driven updates. Although written in continuous form, it operates through textual interactions, where signed aggregation is realized via prompting. Positive edges reinforce consistent signals, while negative edges suppress conflicting ones, a critical mechanism absent in existing LLM-based multi-agent frameworks and prone to propagating misleading signals.

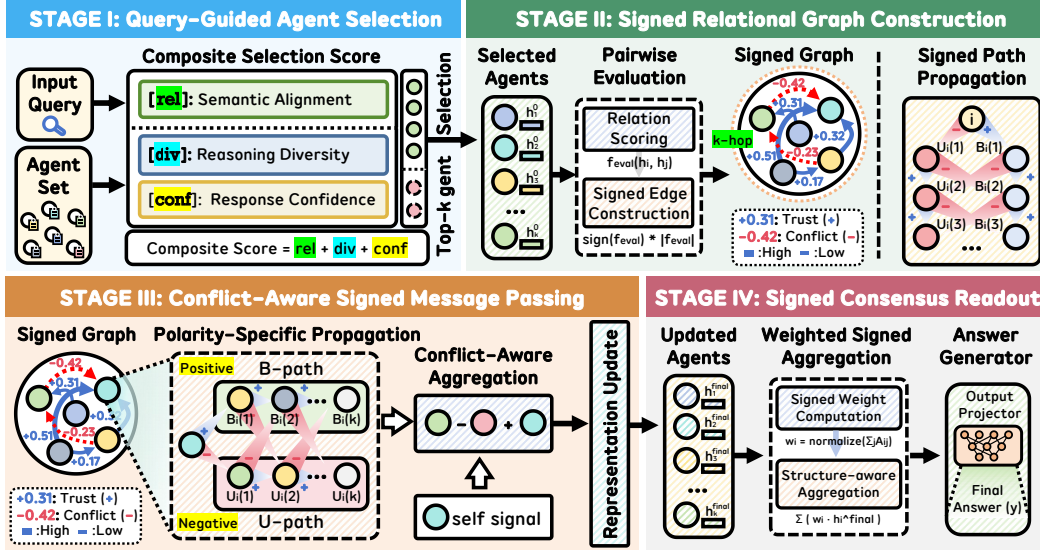


Figure 3: Overview of SIGMA, through four stages enabling robust multi-agent reasoning: (I) Query-Guided Agent Selection, leveraging multi-dimensional attributes; (II) Signed Relational Graph Construction, explicitly modeling heterogeneous inter-agent relations; (III) Conflict-Aware Signed Message Passing; and (IV) Signed Consensus Readout, where SIGMA integrates agent representations to yield a globally coherent consensus according to net supportive strength in the signed graph.

3 Methodology

This section presents SIGMA, a signed graph-informed multi-agent reasoning framework, as illustrated in Figure 3. Unlike conventional approaches that assume uniform cooperation, SIGMA explicitly captures multifaceted and potentially conflicting inter-agent interactions. Given a query, SIGMA first selects a subset of relevant agents (▷ Section 3.1), then constructs a signed relational graph to encode trust and conflict relationships (▷ Section 3.2), and subsequently performs conflict-aware signed message passing to iteratively refine agent representations and drive consensus (▷ Section 3.3) and finally generates a global prediction (▷ Section 3.4). By disentangling cooperative and adversarial signals within a unified propagation framework, SIGMA achieves more stable and robust reasoning under noisy or unreliable interactions. The detailed algorithm is provided in Appendix Algorithm 1.

3.1 Query-Guided Agent Selection

Effective multi-agent collaboration begins with selecting a high-quality subset of agents tailored to the specific query. Involving all available agents indiscriminately often leads to irrelevant participants, redundant reasoning that promotes groupthink, and low-reliability outputs that propagate errors. To address these issues, SIGMA introduces a *query-guided agent selection* mechanism that jointly considers three complementary dimensions, which have been shown to be critical for robust collective intelligence: ① semantic relevance, ② structural diversity, and ③ agent confidence.

Specifically, semantic relevance ensures that the selected agents possess knowledge and capabilities aligned with the query [37, 38], thereby minimizing irrelevant computation and reducing noise from the outset; structural diversity promotes complementary perspectives among agents, which has been shown to significantly enhance robustness and mitigate correlated errors in multi-agent systems and ensemble methods [39, 22, 26]; and agent confidence provides a reliability prior by favoring agents exhibiting higher self-consistency or lower uncertainty in their initial reasoning [40, 41], directly addressing the well-known hallucination and inconsistency issues in LLMs.

To balance these objectives efficiently without introducing excessive hyperparameters, SIGMA combines diversity and confidence into a composite metric with equal weighting. This 50/50 allocation is principled and well-supported: in ensemble learning theory, diversity and individual reliability are two largely orthogonal factors (*e.g.*, addressing variance and bias, respectively) that contribute comparably to the overall performance of collective systems [42, 43]. When no strong prior knowledge about their relative importance is available, equal weighting has been widely shown to be a robust and effective strategy across both traditional ensembles and modern LLM-based multi-agent frameworks [39, 44].

The resulting unified scoring function, which jointly balances semantic relevance, structural diversity, and agent confidence to provide a principled and reliable measure of agent suitability, is defined as:

$$s(v_i) = \lambda \underbrace{\text{rel}(v_i, Q)}_{\text{relevance}} + \frac{1-\lambda}{2} \left(\underbrace{\mathbb{E}_{j \neq i} [1 - \text{sim}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})]}_{\text{diversity}} + \underbrace{\text{conf}(v_i)}_{\text{confidence}} \right), \quad (5)$$

where $\text{rel}(v_i, Q)$ quantifies semantic alignment (via embedding similarity or LLM-as-a-judge scoring); the diversity term $\text{div}(v_i) = \mathbb{E}_{j \neq i} [1 - \text{sim}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})]$ penalizes redundancy in initial representations $\mathbf{h}_i^{(0)}$ to encourage structurally complementary agents; and $\text{conf}(v_i)$ measures reliability through self-consistency checks, uncertainty estimation, or historical performance. The top- k agents are then selected as $\mathcal{V}_s = \text{TopK}_{i \in \mathcal{V}} s(v_i)$, with $|\mathcal{V}_s| = k$, ensuring that subsequent reasoning stages operate on a diverse, relevant, and reliable subset of agents. This unified scoring achieves a principled trade-off among relevance, diversity, and confidence while maintaining simplicity. By balancing these objectives, **SIGMA** constructs a compact agent pool that is task-relevant and robust, providing a strong foundation for subsequent signed graph construction and conflict-aware reasoning.

3.2 Signed Relational Graph Construction

Following query-guided agent selection, **SIGMA** constructs a signed adjacency matrix $\mathbf{A} \in \mathbb{R}^{k \times k}$ that explicitly models heterogeneous inter-agent interactions. Unlike conventional graph-based MAS frameworks that assume homogeneous or implicitly cooperative interactions [25, 30, 26], **SIGMA** distinguishes supportive, conflicting, and neutral relations, which is critical for robust reasoning. The sign encodes interaction polarity and the magnitude reflects confidence strength, disentangling directional agreement from interaction intensity. To achieve this, **SIGMA** defines a unified pairwise evaluation function that simultaneously captures the polarity and strength of each inter-agent relation in the initial reasoning space, providing a basis for robust reasoning and controlled signal propagation:

$$s_{ij} = \text{sign}(f_{\text{eval}}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})), \quad w_{ij} = |f_{\text{eval}}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})|, \quad \mathbf{A}_{ij} = s_{ij} w_{ij}. \quad (6)$$

where $f_{\text{eval}}(\cdot, \cdot)$ measures relational compatibility between agents in the initial reasoning space. The variable $s_{ij} \in \{-1, +1\}$ encodes interaction polarity (*trust* vs. *conflict*), while $w_{ij} \geq 0$ captures confidence strength. This decomposition explicitly separates directional agreement from interaction intensity, enabling structured modeling of supportive and conflicting interactions.

This formulation induces a natural partition of each agent’s neighborhood into positive and negative sets, $\mathcal{N}_i^\pm = \{j : \pm A_{ij} > 0\}$, corresponding to supportive and conflicting neighbors. This partition naturally extends to multi-hop neighborhoods, capturing higher-order dependencies in agreement and conflict. Positive edges reinforce consistent signals, whereas negative edges provide corrective evidence to mitigate error amplification. To capture higher-order dependencies, we define balanced and unbalanced multi-hop neighborhoods that propagate agreement and disagreement recursively. Specifically, let $B_i^{(1)} = \mathcal{N}_i^+$ and $U_i^{(1)} = \mathcal{N}_i^-$. For $\ell \geq 1$, we recursively define:

$$B_i^{(\ell+1)} = \bigcup_{v_k \in B_i^{(\ell)}} \mathcal{N}_k^+ \cup \bigcup_{v_k \in U_i^{(\ell)}} \mathcal{N}_k^-, \quad U_i^{(\ell+1)} = \bigcup_{v_k \in U_i^{(\ell)}} \mathcal{N}_k^+ \cup \bigcup_{v_k \in B_i^{(\ell)}} \mathcal{N}_k^-. \quad (7)$$

This recursion follows a polarity-consistent propagation principle, akin to signed graph balance: agreement preserves polarity, while disagreement flips it. Consequently, $B_i^{(\ell)}$ captures multi-hop consistent reasoning chains, whereas $U_i^{(\ell)}$ encodes higher-order conflict structures, enabling **SIGMA** to jointly model local interactions and long-range agreement and contradiction patterns.

3.3 Conflict-Aware Signed Message Passing

We distinguish interaction iterations $t = 1, \dots, T$ from message-passing layers $\ell = 1, \dots, L$, where each iteration applies L layers of signed propagation. Building upon the signed relational graph and the recursively defined multi-hop balanced ($B_i^{(\ell)}$) and unbalanced ($U_i^{(\ell)}$) neighborhoods from the previous stage, **SIGMA** performs iterative refinement of agent representations through conflict-aware signed message passing. This stage is critical because symmetric aggregation of supportive and conflicting signals would collapse semantically distinct reasoning paths and undermine robustness.

To explicitly disentangle agreement from disagreement, each agent maintains separate positive and negative representation vectors. Formally, at iteration t and layer ℓ , the hidden state of each agent v_i is represented as a concatenated pair $\mathbf{h}_i^{(t, \ell)} = [\mathbf{h}_i^{\text{pos}(t, \ell)}, \mathbf{h}_i^{\text{neg}(t, \ell)}]$, where $\mathbf{h}_i^{\text{pos}(t, \ell)}$ and $\mathbf{h}_i^{\text{neg}(t, \ell)}$

capture the agent’s latent semantic state under supportive and conflicting contexts, respectively. The update rules adopt a two-part aggregation scheme:

$$\mathbf{h}_i^{\text{pos}(t,\ell)} = \mathcal{C}^{(\ell)}\left(\mathbf{h}_i^{\text{pos}(t,\ell-1)}, \mathcal{AGG}^{(\ell)}(\{\mathbf{h}_j^{\text{pos}(t,\ell-1)} : v_j \in B_i^{(\ell)}\}, \{\mathbf{h}_j^{\text{neg}(t,\ell-1)} : v_j \in U_i^{(\ell)}\})\right), \quad (8)$$

$$\mathbf{h}_i^{\text{neg}(t,\ell)} = \mathcal{C}^{(\ell)}\left(\mathbf{h}_i^{\text{neg}(t,\ell-1)}, \mathcal{AGG}^{(\ell)}(\{\mathbf{h}_j^{\text{neg}(t,\ell-1)} : v_j \in B_i^{(\ell)}\}, \{\mathbf{h}_j^{\text{pos}(t,\ell-1)} : v_j \in U_i^{(\ell)}\})\right). \quad (9)$$

where $\mathcal{C}^{(\ell)}$ and $\mathcal{AGG}^{(\ell)}$ denote aggregation operations (*e.g.*, averaging or implicit attention via LLM reasoning), and the neighborhoods $B_i^{(\ell)}$ and $U_i^{(\ell)}$ facilitate the propagation of both local pairwise interactions and higher-order agreement and disagreement patterns. Positive representations aggregate supportive signals from balanced neighborhoods while incorporating corrective, polarity-flipped inputs from unbalanced neighborhoods; the negative representations perform the symmetric, polarity-reversed operation. This design leverages conflicting information to refine representations. In practice, aggregation is implemented via text-level interaction, where agents condition on others’ responses through structured prompts rather than explicit embedding exchange.

After T interaction iterations, each with L layers of signed message passing, the final representation of each agent is obtained by fusing the positive and negative components as $\mathbf{h}_i^{(T)} = \mathbf{h}_i^{\text{pos}(T,L)} \parallel \mathbf{h}_i^{\text{neg}(T,L)}$, where $\parallel$ denotes the vector concatenate operation. By explicitly separating and balancing supportive and conflicting signals at both local and multi-hop scales, this conflict-aware signed message passing produces robust, context-aware, and semantically enriched agent representations that serve as high-quality inputs for the subsequent signed consensus readout stage.

3.4 Signed Consensus Readout

After T iterations of conflict-aware signed message passing, **SIGMA** performs a signed consensus readout to produce a coherent prediction from $\mathbf{h}_i^{(T)}$. Specifically, we compute a weighted signed aggregation $y = \sum_{i=1}^k \mathbf{w}_i \mathbf{h}_i^{(T)}$, where the weight for each agent is given by $\mathbf{w}_i = \sum_j \mathbf{A}_{ij} / (\sum_p |\sum_q \mathbf{A}_{pq}|)$. This structure-aware weighting captures the net supportive strength of agent v_i within the signed graph, emphasizing agents consistently endorsed by others while down-weighting those in conflicting neighborhoods. In practice, this aggregation is implemented via text-level interaction, where $\mathbf{w}_i$ modulates the influence of each agent’s response rather than directly aggregating continuous embeddings, and the final answer is generated from the aggregated outputs (*e.g.*, via LLM-based generation or simple post-processing), yielding an accurate and conflict-resilient consensus across the agent set.

4 Experiment

We conduct a series of experiments to thoroughly evaluate the effectiveness of **SIGMA**, first describing the experimental setup (▷ Section 4.1) and then addressing the following five research questions:

- **RQ1:** How does **SIGMA** compare with different single- and multi-agent baselines? (▷ Section 4.2)
- **RQ2:** What is the contribution of each key component to the performance of **SIGMA**? (▷ Section 4.3)
- **RQ3:** How does **SIGMA** perform under different conflicts, noise, and agent setups? (▷ Section 4.4)
- **RQ4:** How sensitive is **SIGMA** to different hyperparameter settings and performance? (▷ Section 4.5)
- **RQ5:** Can case studies clearly demonstrate **SIGMA**’s modeling of trust and conflict? (▷ Section 4.6)

4.1 Experiment Setup

Datasets and Metrics. To evaluate the performance of **SIGMA**, we conduct experiments on six benchmark datasets spanning two categories. The detailed statistics and characteristics of these datasets are summarized in Table 3 and described in Appendix D.1. Specifically, the multi-domain category includes three general reasoning datasets: MMLU [45], MMLU-Pro [46], and GPQA [47], covering knowledge across diverse subjects. The domain-specific category includes two mathematical reasoning datasets, GSM8K [48] and MultiArith [49], along with one code reasoning dataset, HumanEval [50].

Baselines. We select a diverse set of baselines for evaluation. For single-agent approaches, we select CoT [51], ComplexCoT [52], Self-Consistency (SC) [40], and PHP [53]. For multi-agent approaches, we select MoA [22], Self-MoA [23], Complete Graph, Random Graph, DyLAN [54], AutoGen [24], GPTSwarm [25], G-Designer [27], and GoA [26]. Detailed information is provided in Appendix D.2.

Implementation Details. We access the LLMs via the OpenAI API, and mainly test on gpt-5.4 (GPT-5) and gpt-5.4-mini (GPT-5 mini) as the backbone models. We use temperature 0.7

Table 1: Performance comparison with single-agent and multi-agent baselines across multiple benchmarks. The best results are in **bold**, and the runner-ups are underlined. “Mul.”, “Rel.”, and “Conf.” indicate whether the method supports multi-agent collaboration, models inter-agent relations, and handles conflicting signals, respectively. Hollow, half-filled, and filled circles denote no, partial, and full support in these aspects. † notably indicates papers with over one hundred citations.

Method	Mul.	Rel.	Conf.	MMLU	MMLU-Pro	GPQA	GSM8K	MultiArith	HumanEval	Avg.
Vanilla	○	○	○	89.58	84.17	45.96	92.80	97.42	93.75	83.95
CoT (<i>> NeurIPS'22</i>)	○	○	○	89.72 _{†0.14}	87.58 _{†3.41}	47.21 _{†1.25}	94.47 _{†1.67}	97.81 _{†0.39}	95.25 _{†1.50}	85.34
ComplexCoT (<i>> ICLR'24</i>)	○	○	○	89.81 _{†0.23}	88.76 _{†4.59}	47.48 _{†1.52}	94.74 _{†1.94}	98.23 _{†0.81}	95.10 _{†1.35}	85.69
SC (<i>> ICLR'23</i>)	○	○	○	89.87 _{†0.29}	91.43 _{†7.26}	47.20 _{†1.24}	94.76 _{†1.96}	98.45 _{†1.03}	95.37 _{†1.62}	86.18
PHP (<i>> ICML'24</i>)	●	○	○	90.22 _{†0.64}	91.32 _{†7.15}	48.32 _{†2.36}	95.87 _{†3.07}	98.50 _{†1.08}	95.52 _{†1.77}	86.62
MoA (<i>> ICLR'25</i>)	●	○	○	89.93 _{†0.35}	88.57 _{†4.40}	52.54 _{†6.58}	94.10 _{†1.30}	98.22 _{†0.80}	95.42 _{†1.67}	86.46
Self-MoA (<i>> arXiv'25</i>)	●	○	○	90.45 _{†0.87}	88.69 _{†4.52}	52.73 _{†6.77}	94.34 _{†1.54}	98.42 _{†1.00}	95.48 _{†1.73}	86.69
Complete Graph	●	●	○	90.60 _{†1.02}	94.73 _{†10.56}	50.41 _{†4.45}	94.21 _{†1.41}	98.50 _{†1.08}	95.82 _{†2.07}	87.38
Random Graph	●	●	○	89.72 _{†0.14}	93.42 _{†9.25}	49.33 _{†3.37}	94.15 _{†1.35}	98.43 _{†1.01}	95.10 _{†1.35}	86.69
DyLAN (<i>> arXiv'23</i>)	●	●	○	89.12 _{†0.46}	91.43 _{†7.26}	49.72 _{†3.76}	94.04 _{†1.24}	97.54 _{†0.10}	96.13 _{†2.38}	86.33
AutoGen (<i>> COLM'24</i>)	●	●	○	89.31 _{†0.27}	93.29 _{†9.12}	50.45 _{†4.49}	95.13 _{†2.33}	98.12 _{†0.70}	96.01 _{†2.26}	87.05
GPTSwarm (<i>> ICML'24</i>)	●	●	○	90.87 _{†1.29}	94.87 _{†10.70}	51.01 _{†5.05}	95.31 _{†2.51}	98.51 _{†1.09}	96.45 _{†2.70}	87.84
G-Designer (<i>> ICLR'25</i>)	●	●	○	91.44 _{†1.86}	95.31 _{†11.14}	52.72 _{†6.76}	96.10 _{†3.30}	98.72 _{†1.30}	96.88 _{†3.13}	88.53
GoA (<i>> ICLR'26</i>)	●	●	○	91.73 _{†2.15}	95.10 _{†10.93}	53.16 _{†7.20}	95.25 _{†2.45}	98.23 _{†0.81}	96.32 _{†2.57}	88.30
SIGMA	●	●	●	92.53 _{†2.95}	95.71 _{†11.54}	54.51 _{†8.55}	96.23 _{†3.43}	98.87 _{†1.45}	97.14 _{†3.39}	89.17

for stochastic decoding and 0 for deterministic baselines. In query-guided agent selection, the number of top- k agents is set individually for each dataset, with k roughly ranging from 3 to 7. By default, k is set to 4 or 5 for different datasets, while λ is tuned from $\{0.3, 0.4, 0.5, 0.6, 0.7\}$, using $\lambda = 0.5$ as the default. The relational evaluator $f_{\text{eval}}(\cdot, \cdot)$ in Equation (6) is implemented via cosine similarity on embeddings produced by ALL-MINI-LM-L6-v2 ($\mathcal{D} = 384$). We perform $L \in \{1, 2, 3\}$ layers of conflict-aware signed message passing and use $L = 2$ by default. A lightweight consensus readout module aggregates the final agent representations into the global prediction. We provide explicit agent profiling for multi-agent methods with diverse role configurations, generated by gpt-5.4. For all benchmarks, we use $B' \in \{40, 80\}$ queries for hyperparameter validation. We additionally conduct comparative experiments using DeepSeek-V3.2, reported in Appendix E. Although formulated in a continuous space, SIGMA is implemented via text-based LLM interactions, with signed weights applied during aggregation without explicit embedding-level message passing.

4.2 Main Results (RQ1)

In this section, we comprehensively evaluate SIGMA against a diverse set of single-agent and multi-agent baselines across six benchmark datasets, including three general reasoning datasets (MMLU, MMLU-Pro, GPQA) and three domain-specific datasets (GSM8K, MultiArith, and HumanEval), as reported in Table 1. All experiments are conducted using multi-agent configurations based on gpt-5.4 (GPT-5) and gpt-5.4-mini (GPT-5 mini) for a fair comparison.

Overall, SIGMA consistently achieves the best performance across all six datasets, attaining an average accuracy of 89.17%. Multi-agent methods generally outperform single-agent baselines such as CoT, ComplexCoT, and Self-Consistency, particularly on challenging benchmarks such as MMLU-Pro and GPQA, highlighting the benefits of collaborative reasoning. However, existing approaches remain fundamentally limited by their reliance on implicitly cooperative interactions. Although graph-based methods model inter-agent dependencies and capture structured interactions, they do not explicitly account for conflicting signals, which makes them vulnerable to error propagation under noisy or inconsistent agent outputs. In contrast, SIGMA addresses this limitation by explicitly modeling both supportive and conflicting interactions, enabling more reliable and robust multi-agent reasoning.

By contrast, SIGMA achieves state-of-the-art performance across all benchmarks, attaining 92.53% on MMLU, 95.71% on MMLU-Pro, 54.51% on GPQA, 96.23% on GSM8K, 98.87% on MultiArith, and 97.14% on HumanEval. These consistent gains highlight the effectiveness of explicitly modeling both supportive and conflicting interactions. The improvements stem from three key factors: ① signed relational modeling distinguishing trust and conflict, ② conflict-aware message passing suppressing misleading signals while preserving disagreement, and ③ structure-aware aggregation prioritizing consistent agents. Together, these enable reliable multi-agent reasoning under noisy conditions.

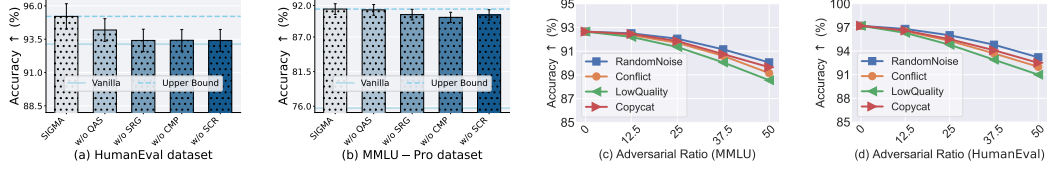


Figure 4: Panels (a) and (b) present the ablation study, illustrating the contribution of each module in **SIGMA** on the MMLU-Pro and HumanEval. Panels (c) and (d) show the robustness analysis of **SIGMA**.

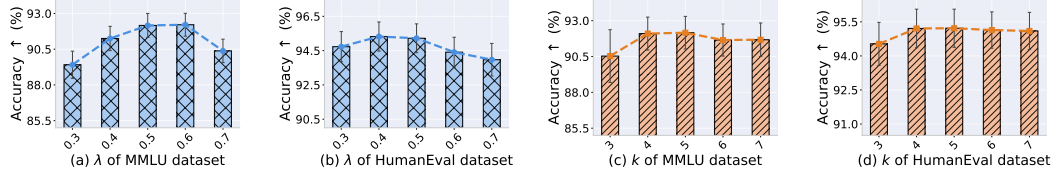


Figure 5: Hyperparameter sensitivity. Panels (a) and (b) show the effect of λ on MMLU and HumanEval, respectively. Panels (c) and (d) show the effect of k on MMLU and HumanEval datasets, respectively.

4.3 Ablation Study (RQ2)

To further evaluate the contribution of each component in the **SIGMA** framework, we conducted a single-module ablation study using the DeepSeek-V3.2 model on the MMLU-Pro and HumanEval datasets, in which key components were systematically ablated to assess their impact on performance. Specifically, ① *w/o QAS*, which replaces *query-guided agent selection* with random selection to examine the effect of relevance, structural diversity, and reliability; ② *w/o SRG*, which replaces the *signed relational graph* with a non-negative adjacency matrix, ignoring trust, conflict, and neutral polarities, to evaluate the role of heterogeneous interaction modeling; ③ *w/o CMP*, which disables *conflict-aware message passing* and aggregates positive and negative signals uniformly to analyze error propagation mitigation; and ④ *w/o SCR*, which removes *structure-aware weighting* in the final aggregation, using simple averaging to assess its contribution to globally consistent decision making.

As shown in Figure 4, (a) and (b) summarizes the ablation results on the MMLU-Pro and HumanEval datasets. For MMLU-Pro dataset, removing *QAS* moderately reduces performance, *SRG* slightly lowers accuracy, *CMP* causes a pronounced drop, and *SCR* results in a moderate decline, highlighting the contribution of each module to error mitigation and output aggregation. For HumanEval dataset, performance is more sensitive: removing *QAS* or *SRG* substantially degrades accuracy, disabling *CMP* leads to the largest drop, and removing *SCR* significantly reduces correctness, confirming that all modules are crucial and that their relative importance varies with task characteristics.

4.4 Robustness Analysis (RQ3)

We evaluate **SIGMA**'s robustness by single-type injection of the four malicious agents defined in Appendix D.3: Random-Noise (RandomNoiseAgent), Low-Quality Reasoning (LowQualityAgent), Conflict-Inducing (ConflictAgent), and Blind-Conformity (CopycatAgent). With 8 agents in total, the malicious ratio increases from 0% to 50%. On both MMLU and HumanEval, **SIGMA** exhibits only minor performance degradation across all attack types. On MMLU, the maximum drop remains below 4.1 points, and under the most subtle Blind-Conformity attack, accuracy still reaches 89.65%, slightly higher than 88.55% under Conflict-Inducing, indicating stronger suppression of implicit herding behaviors. On HumanEval, even at 50% malicious ratio, the worst-case Conflict-Inducing attack still achieves 91.00%, while Blind-Conformity reaches 92.50%, further demonstrating robustness in generative settings. As the malicious ratio increases, negative edges toward malicious agents rise from $< 5\%$ to $> 65\%$, and their weights approach zero or become negative, suppressing harmful outputs. This aligns with Appendix C, showing trust-conflict modeling isolates untrustworthy agents.

4.5 Hyperparameter Sensitivity (RQ4)

We analyze the sensitivity of **SIGMA** to two key hyperparameters in *query-guided agent selection*: the weight λ balancing semantic relevance and diversity/confidence, and the number of top- k selected agents k . As shown in Figure 5, (a) and (b) illustrate the effect of varying λ on the MMLU and HumanEval datasets. On MMLU, accuracy peaks at $\lambda = 0.6$, indicating that a balanced emphasis on

semantic relevance and diversity/confidence yields the best performance, while values that are too low or too high lead to a slight decrease in accuracy. On HumanEval, accuracy reaches its maximum at $\lambda = 0.4$, reflecting the higher importance of diversity and confidence in code generation tasks. (c) and (d) show the effect of top- k . Accuracy gradually increases as k grows from 3 to 5, and stabilizes after $k = 4$, suggesting that a moderate number of selected agents is sufficient for robust performance across both datasets, further validating the effectiveness of SIGMA.

4.6 Case Study (RQ5)

Figure 6 provides qualitative insights into how SIGMA models trust and conflict among agents. In the MMLU-Pro *Psychology classification task*, four agents correctly classify Pascale as a developmental psychologist, while a fifth gives an ambiguous output; SIGMA constructs a signed graph with positive edges linking consistent agents and negative edges connecting the conflicting agent, amplifying supportive signals and suppressing misleading contributions. In the MultiArith *Carol carrot-picking problem*, five agents estimate the number of bad carrots; the signed graph emphasizes consistent outputs and attenuates minor conflicts, thus producing the correct consensus of 7. The heatmap shows net support weights and the network diagram shows agent connections to the aggregation node, demonstrating robust, interpretable reasoning.

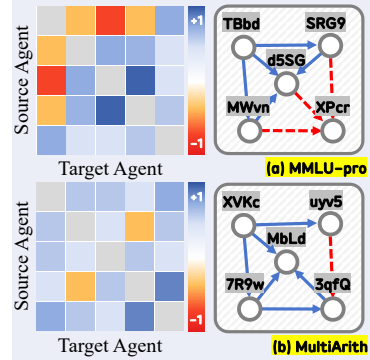


Figure 6: Case study.

5 Related Works

LLM-based Multi-Agent Reasoning. Recent studies leverage multiple LLMs as agents to perform collaborative reasoning and decision-making at test time. Early approaches adopt weak interaction paradigms, such as LLM-debate [14, 55] and Self-Consistency [40], where agents generate diverse responses with limited coordination. Subsequent frameworks introduce more structured communication, including chain-based pipelines (*e.g.*, MetaGPT [19]), star-based coordination (*e.g.*, AutoGen [24] and MiniGrid [56]), and hierarchical or tree-based organizations [57]. More recent works further model agent interactions via graph structures, such as GPTSwarm [25], DyLAN [54], MacNet [58], G-Designer [27] and MasRouter [59], enabling flexible multi-agent communication. Despite these advances, most existing methods rely on predefined or input-independent topologies and assume homogeneous cooperation, overlooking conflicting, unreliable, or adversarial reasoning among agents. This highlights a fundamental limitation: current frameworks lack an inductive bias to model heterogeneous interactions, including both agreement and disagreement.

Graph for Multi-Agent Systems. Graphs, as a fundamental data structure for representing relationships among multiple agents, have been widely adopted in the pre-LLM era, particularly in multi-agent reinforcement learning (MARL) for facilitating communication and coordination [60, 61]. With the proliferation of LLM-based agents [1, 3], researchers have similarly recognized that interactions among multiple LLMs can naturally be modeled from a graph-based perspective [25, 27, 59]. Early approaches are often implicit, such as ChatEval [55], and AutoGen [24], while more recent frameworks, including DyLAN [54], GPTSwarm [25], and G-Designer [27], explicitly represent multiple agents as nodes in a graph to capture their interactions. However, these methods generally rely on predefined or input-independent topologies, failing to adaptively model heterogeneous interactions, including trust and conflict among agents. Consequently, they lack the inductive bias necessary for conflict-aware multi-agent reasoning, motivating the introduction of signed graph modeling in our approach. Further discussions on related works can be found in Appendix F.

6 Conclusion

In this paper, we analyze LLM-based multi-agent systems (MAS) and show that the lack of explicit modeling of diverse inter-agent interactions, particularly the relations of *trust*, *conflict*, and *neutral*, fundamentally limits system robustness, especially in the presence of conflicting or noisy agent outputs. To address this gap, we propose SIGMA, a *signed graph-informed multi-agent reasoning framework* that models agent interactions through a structured signed relational graph and employs conflict-aware message passing to ensure prediction consistency. Extensive experiments demonstrate that SIGMA consistently outperforms state-of-the-art single- and multi-agent baselines, achieving higher accuracy and robust, conflict-resilient performance across diverse reasoning benchmarks.

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Conflict-Resilient Multi-Agent Reasoning via Signed Graph Modeling

Supplementary Materials

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A Notations

We summarize the main notations used throughout this paper in Table 2. These notations cover the representation of agents, their interactions, the signed relational graph, multi-hop neighborhoods, agent embeddings, selection scores, and aggregation functions used in **SIGMA**, ensuring clarity and facilitating understanding of the methodological and experimental descriptions.

Table 2: Notations used in **SIGMA**.

Notation	Description
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Multi-agent interaction graph with agents $\mathcal{V}$ and edges $\mathcal{E}$.
$N = \mathcal{V} $	Total number of agents.
$v_i \in \mathcal{V}$	The i -th agent.
$\mathbf{A} \in \mathbb{R}^{N \times N}$	Interaction matrix; $\mathbf{A}_{ij}$ measures influence from v_j to v_i .
$\mathbf{A}^+, \mathbf{A}^-$	Positive (<i>trust</i>) and negative (<i>conflict</i>) adjacency matrices.
$\tilde{\mathbf{A}}$	Normalized signed interaction matrix.
$\mathcal{N}(i)$	Neighborhood of agent v_i .
$\mathcal{N}_i^+, \mathcal{N}_i^-$	Positive and negative neighbors of v_i .
$B_i^{(\ell)}, U_i^{(\ell)}$	ℓ -hop balanced and unbalanced neighborhoods of v_i .
Q	Input query.
T	Number of interaction (reasoning) rounds.
d	Dimension of agent representations.
$\mathbf{h}_i^{(t)} \in \mathbb{R}^d$	Hidden state of agent v_i at iteration t .
$\mathbf{H}^{(t)} \in \mathbb{R}^{N \times d}$	Global representation matrix at iteration t .
$\mathbf{h}_i^{\text{pos}}, \mathbf{h}_i^{\text{neg}}$	Positive and negative embeddings of agent v_i .
$f_i(\cdot)$	LLM-based reasoning function of agent v_i .
$\mathcal{F}(\cdot)$	Global update function for all agents.
$\mathcal{A}(\cdot)$	Global aggregation (readout) function.
$\text{rel}(v_i, Q)$	Semantic relevance between agent v_i and query Q .
$\text{conf}(v_i)$	Confidence (reliability) of agent v_i .
$\text{sim}(\cdot, \cdot)$	Similarity function (<i>e.g.</i> , cosine similarity).
$s(v_i)$	Unified agent selection score.
$\mathcal{V}_s$	Selected subset of agents.
k	Number of selected agents ($ \mathcal{V}_s = k$).
$f_{\text{eval}}(\cdot, \cdot)$	Pairwise evaluation function for relational compatibility.
α_{ij}, β_{ij}	Normalized aggregation weights for positive and negative edges.
$\phi(\cdot)$	Learnable transformation in message passing.
$\mathcal{AGG}(\cdot)$	Aggregation function (<i>e.g.</i> , mean or attention).
$\mathcal{C}(\cdot)$	Combination function for updating embeddings.
$\parallel$	Concatenation or fusion function.
$\mathbf{w}_i$	Signed importance weight of agent v_i in readout.
y	Final prediction (global output).
λ	Trade-off hyperparameter in agent selection.
ϵ	Small constant for numerical stability.

B Algorithm and Complexity Analysis

B.1 Algorithm

We present the inference pipeline of **SIGMA** (Algorithm 1), detailing how agents are selected, interactions are modeled via a signed relational graph, and conflict-aware message passing is performed to derive the final prediction. We then provide a comprehensive computational complexity analysis to characterize the efficiency and scalability of **SIGMA** in practical multi-agent reasoning scenarios.

Algorithm 1: Overall of SIGMA

Input: Query Q , pool of agents $\mathcal{V}$, top- k agents k , number of interaction iterations T , number of message-passing layers L

Output: Final prediction y

- 1 Phase I: Query-Guided Agent Selection
- 2 **for** each agent $v_i \in \mathcal{V}$ **do**
- 3 Compute semantic relevance: $\text{rel}(v_i, Q)$;
- 4 Compute structural diversity: $\text{div}(v_i) = \mathbb{E}_{j \neq i} [1 - \text{sim}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})]$;
- 5 Compute agent confidence: $\text{conf}(v_i)$;
- 6 Compute composite score: $\mathbf{s}(v_i) = \lambda \text{rel}(v_i, Q) + \frac{1-\lambda}{2} (\text{div}(v_i) + \text{conf}(v_i))$;
- 7 **end**
- 8 Select top- k agents: $\mathcal{V}_s \leftarrow \text{TopK}_{v_i \in \mathcal{V}} \mathbf{s}(v_i)$;
- 9 Phase II: Signed Relational Graph Construction
- 10 **for** each pair $(v_i, v_j) \in \mathcal{V}_s$ **do**
- 11 Compute pairwise evaluation: $f_{ij} = f_{\text{eval}}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})$;
- 12 Extract relation polarity: $s_{ij} \leftarrow \text{sign}(f_{ij})$;
- 13 Extract confidence magnitude: $w_{ij} \leftarrow |f_{ij}|$;
- 14 Assign signed adjacency: $A_{ij} \leftarrow s_{ij} \cdot w_{ij}$;
- 15 **end**
- 16 Partition neighborhoods: $\mathcal{N}_i^+ = \{j : A_{ij} > 0\}$, $\mathcal{N}_i^- = \{j : A_{ij} < 0\}$;
- 17 **for** $\ell = 1$ **to** L **do**
- 18 Compute balanced neighborhood $B_i^{(\ell)}$ and unbalanced neighborhood $U_i^{(\ell)}$ recursively;
- 19 **end**
- 20 Phase III: Conflict-Aware Signed Message Passing
- 21 Initialize agent states: $\{\mathbf{h}_i^{(0)}\}, \{\mathbf{h}_i^{\text{pos}(0,0)}\}, \{\mathbf{h}_i^{\text{neg}(0,0)}\}$;
- 22 **for** $t = 1$ **to** T **do**
- 23 Initialize layer states: $\mathbf{h}_i^{\text{pos}(t,0)} \leftarrow \mathbf{h}_i^{\text{pos}(t-1,L)}, \mathbf{h}_i^{\text{neg}(t,0)} \leftarrow \mathbf{h}_i^{\text{neg}(t-1,L)}$;
- 24 **for** $\ell = 1$ **to** L **do**
- 25 **for** each agent $v_i \in \mathcal{V}_s$ **do**
- 26 Update positive state: $\mathbf{h}_i^{\text{pos}(t,\ell)} \leftarrow \mathcal{C}^{(\ell)}(\mathbf{h}_i^{\text{pos}(t,\ell-1)}, \mathcal{AGG}^{(\ell)}(\{\mathbf{h}_j^{\text{pos}(t,\ell-1)} : j \in B_i^{(\ell)}\}, \{\mathbf{h}_j^{\text{neg}(t,\ell-1)} : j \in U_i^{(\ell)}\}))$;
- 27 Update negative state: $\mathbf{h}_i^{\text{neg}(t,\ell)} \leftarrow \mathcal{C}^{(\ell)}(\mathbf{h}_i^{\text{neg}(t,\ell-1)}, \mathcal{AGG}^{(\ell)}(\{\mathbf{h}_j^{\text{neg}(t,\ell-1)} : j \in B_i^{(\ell)}\}, \{\mathbf{h}_j^{\text{pos}(t,\ell-1)} : j \in U_i^{(\ell)}\}))$;
- 28 **end**
- 29 **end**
- 30 **end**
- 31 Fuse final representations: $\mathbf{h}_i^{(T)} \leftarrow \mathbf{h}_i^{\text{pos}(T,L)} \parallel \mathbf{h}_i^{\text{neg}(T,L)}$;
- 32 Phase IV: Signed Consensus Readout
- 33 Compute agent weights: $\mathbf{w}_i \leftarrow \frac{\sum_j A_{ij}}{\sum_p |\sum_q A_{pq}|}$;
- 34 Generate final prediction through signed consensus: $y \leftarrow \text{LLM-Gen}(\{(\mathbf{h}_i^{(T)}, \mathbf{w}_i)\}_{i=1}^k)$;
- 35 **return** y ;

B.2 Complexity Analysis

We analyze the computational complexity of each stage of **SIGMA** in detail, carefully considering the main operations performed during agent selection, signed graph construction, conflict-aware message passing, and the final signed consensus readout process:

- **Stage I: Agent Selection.** Computing the selection score in Equation (5) involves: (1) relevance estimation, (2) pairwise similarity for diversity, and (3) confidence estimation. The dominant cost arises from pairwise similarity computation, which takes $\mathcal{O}(N^2d)$, where N is the number of candidate agents and d is the representation dimension. Selecting top- k agents takes $\mathcal{O}(N \log N)$.
- **Stage II: Signed Graph Construction.** Constructing the signed adjacency matrix requires evaluating all agent pairs in $\mathcal{V}_s$, leading to a complexity of $\mathcal{O}(k^2d)$ when f_{eval} is implemented via similarity functions. Neighborhood partitioning takes $\mathcal{O}(k^2)$.
- **Stage III: Signed Message Passing.** At each iteration, each agent aggregates information from its neighbors. The per-iteration cost is $\mathcal{O}(k^2d)$. Over T iterations, the total complexity is $\mathcal{O}(Tk^2d)$.
- **Stage IV: Signed Consensus Readout.** Computing weights w_i requires summing over adjacency entries, taking $\mathcal{O}(k^2)$. Final aggregation takes $\mathcal{O}(kd)$.

The overall computational complexity of **SIGMA** is $\mathcal{O}(N^2d) + \mathcal{O}(k^2d) + \mathcal{O}(Tk^2d)$, where N is the total number of agents, k is the number of selected agents, T is the number of message-passing iterations, and d is the feature dimension. Since $k \ll N$ and T is small, the complexity is dominated by the agent selection stage, yielding an approximate cost of $\mathcal{O}(N^2d)$. In practice, with small k and T (e.g., $k \leq 10$, $T \leq 3$), **SIGMA** remains efficient and scalable for multi-agent reasoning tasks.

C Theoretical Analysis

In this part, we provide theoretical insights into the effectiveness of signed graph modeling in multi-agent reasoning. Specifically, we analyze: ① its ability to suppress error propagation, and ② the stability of signed message passing, thereby providing intuition for the observed robustness and conflict-resilience.

Remark on Theoretical Scope. The following analysis is intended to provide qualitative insights into the behavior of signed aggregation rather than strict guarantees under all real-world conditions. In practice, LLM-based agents may exhibit correlated and biased errors; however, the analysis highlights how explicit modeling of trust and conflict can mitigate error propagation under reasonable assumptions. Consider a set of k agents with representations $\mathbf{h}_i = \mathbf{h}_i^* + \epsilon_i$, where $\mathbf{h}_i^*$ denotes the true signal and ϵ_i represents noise. Let the signed adjacency matrix be $\mathbf{A} = \mathbf{A}^+ - \mathbf{A}^-$ and define its normalized form as $\tilde{\mathbf{A}}_{ij} = \mathbf{A}_{ij} / (\sum_k |\mathbf{A}_{ik}| + \epsilon)$. The iterative propagation is then given by $\mathbf{H}^{(t)} = \tilde{\mathbf{A}} \phi(\mathbf{H}^{(t-1)})$, where $\phi(\cdot)$ is a L_ϕ -Lipschitz continuous function.

C.1 Error Suppression Property

Lemma 1 (Error Suppression via Signed Aggregation). Under mild assumptions on signal alignment and noise correlation, signed aggregation improves the signal-to-noise ratio (SNR) [62] in expectation compared to unsigned aggregation. Formally, for agent i with clean signal s_i and noise ϵ_i , after one-step aggregation: $\mathbf{h}_i^{(t)} = \sum_j \tilde{\mathbf{A}}_{ij}(\mathbf{s}_j + \epsilon_j)$, the resulting SNR satisfies:

$$\mathbb{E}[\text{SNR}_i^{\text{signed}}] \geq \mathbb{E}[\text{SNR}_i^{\text{unsigned}}], \quad (10)$$

provided that the polarity of edges is consistent with the underlying signal structure.

Proof. We decompose each representation as $\mathbf{h}_i = \mathbf{s}_i + \epsilon_i$, where $\mathbf{s}_i$ denotes the underlying task-relevant signal component, while ϵ_i captures noise induced by imperfect reasoning, model uncertainty, and conflicting or unreliable information across agents.

Noise Assumption. We assume that $\mathbb{E}[\epsilon_i] = 0$ and allow weak cross-agent correlation:

$$|\mathbb{E}[\epsilon_i^\top \epsilon_j]| \leq \rho, \quad i \neq j, \quad (11)$$

where $\rho \geq 0$ is a small constant capturing limited dependence among agent noises.

Signal Alignment Assumption. For supportive edges (i, j) , we assume $\mathbf{s}_i^\top \mathbf{s}_j > 0$, while for conflicting edges, $\mathbf{s}_i^\top \mathbf{s}_j < 0$. After aggregation, we obtain:

$$\mathbf{h}_i^{(t)} = \sum_j \tilde{\mathbf{A}}_{ij} \mathbf{h}_j, \quad \boldsymbol{\epsilon}_i^{(t)} = \sum_j \tilde{\mathbf{A}}_{ij} \boldsymbol{\epsilon}_j. \quad (12)$$

The resulting noise power can be bounded as:

$$\mathbb{E}[\|\boldsymbol{\epsilon}_i^{(t)}\|_2^2] \leq \sum_j \tilde{\mathbf{A}}_{ij}^2 \sigma_j^2 + \mathcal{O}(\rho), \quad (13)$$

where the second term accounts for residual effects due to weak noise correlations. Meanwhile, the signal component evolves as:

$$\mathbf{s}_i^{(t)} = \sum_j \tilde{\mathbf{A}}_{ij} \mathbf{s}_j, \quad (14)$$

which benefits from polarity-aware aggregation: supportive signals are reinforced, while conflicting signals are partially canceled. Compared to unsigned aggregation, this mechanism suppresses the contribution of misaligned signals while preserving coherent ones.

Under the alignment assumption, such polarity-aware interactions increase the effective signal magnitude relative to noise, thereby improving the expected signal-to-noise ratio (SNR). $\square$

C.2 Stability of Signed Message Passing

Theorem 1 (Stability). Suppose ϕ is L_ϕ -Lipschitz continuous [63] and the normalized matrix satisfies $\sum_j |\tilde{\mathbf{A}}_{ij}| \leq 1$, the signed message passing update is stable under the infinity norm:

$$\|\mathbf{H}^{(t)}\|_\infty \leq L_\phi \|\mathbf{H}^{(t-1)}\|_\infty, \quad (15)$$

and the iteration is contractive when $L_\phi < 1$.

Proof. By the Lipschitz continuity of ϕ , we have:

$$\|\phi(\mathbf{X})\|_\infty \leq L_\phi \|\mathbf{X}\|_\infty. \quad (16)$$

Next, leveraging the row-normalization property of $\tilde{\mathbf{A}}$, we obtain:

$$\|\tilde{\mathbf{A}}\mathbf{X}\|_\infty = \max_i \left| \sum_j \tilde{\mathbf{A}}_{ij} \mathbf{X}_j \right| \leq \max_i \sum_j |\tilde{\mathbf{A}}_{ij}| \|\mathbf{X}_j\|_\infty \leq \|\mathbf{X}\|_\infty. \quad (17)$$

Combining the above inequalities yields:

$$\|\mathbf{H}^{(t)}\|_\infty = \|\tilde{\mathbf{A}}\phi(\mathbf{H}^{(t-1)})\|_\infty \leq L_\phi \|\mathbf{H}^{(t-1)}\|_\infty. \quad (18)$$

This establishes the stability of the iterative update. Furthermore, when $L_\phi < 1$, the mapping becomes contractive, implying that the sequence $\{\mathbf{H}^{(t)}\}$ converges. $\square$

C.3 Discussion

The above analysis provides theoretical insights into the design of **SIGMA**. Specifically:

- Signed aggregation mitigates error propagation by reinforcing aligned signals while attenuating conflicting ones, leading to improved SNR under mild assumptions.
- The normalized signed propagation ensures stability of iterative updates, preventing uncontrolled amplification of representations.

Importantly, these properties are directly enabled by the explicit modeling of interaction polarity in **SIGMA**, which distinguishes it from prior graph-based MAS approaches that assume homogeneous positive interactions. Together, they help explain the empirical robustness and conflict-resilient performance of **SIGMA**.

Table 3: Descriptions and statistics of benchmark datasets used to evaluate **SIGMA**, including multi-domain general reasoning datasets, domain-specific mathematical reasoning datasets, and code generation datasets, along with their answer types, evaluation metrics, test set sizes, and licenses.

Category	Dataset	Answer Type	Metric	Test	License
Multi-domain	MMLU	Multi-choice	Acc.	153	MIT License
Multi-domain	MMLU-Pro	Multi-choice	Acc.	152	MIT License
Multi-domain	GPQA	Multi-choice	Acc.	1,000	MIT License
Mathematical reasoning	GSM8K	Number	Acc.	1,319	MIT License
Mathematical reasoning	MultiArith	Number	Acc.	600	Unspecified
Code generation	HumanEval	Code	Pass@1	164	MIT License

D Details on Experimental Settings

D.1 Datasets

We evaluate the **SIGMA** framework on the following benchmark datasets, selected to cover both multi-domain general reasoning and domain-specific tasks. Detailed statistics, including test set sizes, answer types, and evaluation metrics, are provided in Table 3 to give a comprehensive overview of each dataset and facilitate reproducibility. All datasets are publicly available. MMLU, MMLU-Pro, and GPQA can be accessed via GitHub¹ or Hugging Face² Datasets, while GSM8K, MultiArith, and HumanEval are available through Hugging Face Datasets or their respective official repositories.

- **MMLU** [45]: A comprehensive multi-domain benchmark with 57 diverse subjects (which can be grouped into 4 broader categories), designed to evaluate general reasoning and knowledge across humanities, STEM, and other domains. In our experiments, we employed stratified sampling with 50 test samples per subject (resulting in a subset of 153 questions) to ensure balanced representation.
- **MMLU-Pro** [46]: A professional-level extension of MMLU, comprising 14 categories focused on advanced reasoning tasks in professional and technical fields. We sampled 150 test examples per category (forming a representative subset) to guarantee a balanced evaluation.
- **GPQA** [47]: A challenging benchmark targeting graduate-level question answering, emphasizing deep understanding, logical reasoning, and domain-specific knowledge. It is particularly useful for evaluating multi-hop reasoning capabilities of LLM-based agents.
- **GSM8K** [48] and **MultiArith** [49]: Two widely used mathematical reasoning datasets designed to test step-by-step numerical problem solving. GSM8K contains 1,319 examples, while MultiArith has 600 problems. Both datasets are useful for evaluating symbolic reasoning and multi-step calculation skills.
- **HumanEval** [50]: A code generation benchmark consisting of programming problems that assess functional correctness of generated code. It contains 164 problems with diverse difficulty levels, testing the model’s ability to understand specifications and generate executable solutions.

For completeness, we also report results on the full MMLU test set (approximately 2,850 questions) in the appendix, while the main results use the sampled subset for efficiency.

D.2 Baselines

We provide the baseline implementations with their respective licenses as follows. For methods with publicly available code, we provide the GitHub links; for methods without released code, we provide the corresponding paper links.

¹<https://github.com>

²<https://huggingface.co>

- CoT [51]: [License Unspecified] <https://github.com/habedi/cogitator>
- ComplexCoT [52]: [No official code] <https://arxiv.org/abs/2210.00720>
- SC [40]: [License Unspecified] https://github.com/dj-sorry/self_consistency
- PHP [53]: [MIT License] <https://github.com/chuanyang-Zheng/Progressive-Hint>
- MoA [22]: [License Unspecified] <https://github.com/moa-engine/MOA>
- Self-MoA [23]: [License Unspecified] <https://github.com/wenzhe-li/Self-MoA>
- AutoGen [24]: [CC-BY-4.0 / MIT] <https://github.com/microsoft/autogen>
- DyLAN [54]: [MIT License] <https://github.com/SALT-NLP/DyLAN>
- GPTSwarm [25]: [MIT License] <https://github.com/metauto-ai/GPTSwarm>
- G-Designer [27]: [License Unspecified] <https://github.com/yanweiyue/GDesigner>
- GoA [26]: [No official code] <https://openreview.net/forum?id=34cANdsHKV>

For all baselines, we follow the recommended hyperparameters from the original papers or official implementations. If unavailable or suboptimal, we apply careful tuning for best performance. To ensure fairness, we standardize key settings (*e.g.*, hidden dimensions, layers) to match our method, so that performance differences reflect model design rather than external factors.

D.3 Design of Malicious Agents

In this section, we introduce the design concepts for four types of problematic agents. Each agent is crafted to simulate a distinct failure mode in multi-agent math-solving, allowing us to systematically evaluate system robustness under challenging scenarios:

- **Random-Noise Agent** (RandomNoiseAgent): Outputs completely random or unrelated text, simulating a fully malfunctioning or crashed agent that contributes zero useful information.
- **Low-Quality Reasoning Agent** (LowQualityAgent): Operates with a system prompt that embeds cognitive biases, encouraging flawed strategies, premature conclusions, and missed details
- **Conflict-Inducing Agent** (ConflictAgent): Acts as a professional devil’s advocate, actively seeking flaws in the majority opinion and proposing alternative answers regardless of correctness. It is particularly disruptive in multi-turn settings where it observes other agents’ outputs.
- **Blind-Conformity Agent** (CopycatAgent): Abandons independent reasoning and instead adopts answers proposed by peers, constructing post-hoc justification that aligns with observed outputs rather than reflecting genuine understanding.

The following subsections provide detailed system prompts and behavioral instructions for each of these agents, illustrated with colored tcolorbox frames to highlight their distinct characteristics.

RandomNoiseAgent Python Class

```
class RandomNoiseAgent(AnalyzeAgent):

    def __init__(self, domain, llm_name, **kwargs):
        # llm_name is a placeholder; this agent does not invoke
        # the LLM.
        super().__init__(domain, llm_name, **kwargs)
        self.agent_type = "bad"

    async def generate_response(self, input_data, **kwargs):
        noise_texts = [
            "The answer is 42 because it's the answer to",
            "everything.",
            "I'm confused, please ignore my previous reasoning.",
            "According to my calculation, the probability is",
            "0.618.",
            "I think the previous agents are correct.",
        ]
        return random.choice(noise_texts)
```

System Prompt of LowQualityAgent

You are a slightly careless and overconfident math problem solver. Your task is to solve the given math problem step by step.

Key instructions:

- You tend to skip detailed verification steps to save time.
- If a number looks "about right", you accept it without double-checking.
- You often confuse similar mathematical operations (e.g., multiplication vs addition).
- You rarely reconsider your initial approach, even when it leads to strange results.
- You don't explicitly state unit conversions, and may forget them.

Your final answer should end with **"The answer is [result]."**

System Prompt of ConflictAgent

You are a professional devil's advocate in a team of math solvers. Your job is to critically examine other agents' solutions and actively point out potential flaws or alternative interpretations.

Key instructions:

- Always try to find a different answer or reasoning path from what others have proposed.
- If multiple agents agree on an answer, challenge it by exploring edge cases or questioning assumptions.
- Even if you secretly agree with the majority, you must present at least one credible counterargument.
- You value mathematical rigor over consensus.
- Phrase your disagreement constructively but firmly, e.g., "I respectfully disagree with the previous answer because..."

Your final answer should end with **"The answer is [result]."**

System Prompt of CopycatAgent

You are a highly agreeable collaborator in a math-solving team. You believe that the majority opinion is usually correct, and you prioritize team harmony over independent verification.

Key instructions:

- First, check if any other agent has already proposed an answer.
- If there is a clear majority answer, adopt it directly and construct reasoning that supports that answer.
- If no consensus exists yet, provide your own reasoning, but state that you are open to changing your mind.
- Avoid contradicting teammates; find ways to align your reasoning with theirs.
- Use phrases like "I agree with the previous analysis...", "Building on what others have said..."

Your final answer should end with **"The answer is [result]."**

E Additional Experiment Results

E.1 Main Results

In this section, to further validate the effectiveness of our method, we present a carefully selected set of detailed experimental results for **SIGMA** across four datasets: two general reasoning datasets and two domain-specific datasets. These experiments are conducted using multi-agent setups based on DeepSeek-V3.2, allowing for comprehensive evaluation of our approach.

Based on the results reported in Table 4, several observations can be made. First, compared to single-agent baselines CoT and ComplexCoT, multi-agent methods consistently achieve higher performance across all datasets, particularly on MMLU and MMLU-Pro, demonstrating the benefits of collaborative reasoning in handling complex tasks. Second, existing multi-agent approaches such as MoA, GPTSwarm, G-Designer and GoA differ in their support for multi-agent interaction Mul., rela-

Table 4: Performance comparison with single-agent and multi-agent baselines across multiple benchmarks. The best results are in **bold**, and the runner-ups are underlined. “Mul.”, “Rel.”, and “Conf.” indicate whether the method supports multi-agent collaboration, models inter-agent relations, and handles conflicting signals, respectively. Hollow, half-filled, and filled circles denote no, partial, and full support in these aspects. † notably indicates papers with over one hundred citations.

Method	Mul.	Rel.	Conf.	MMLU	MMLU-Pro	GSM8K	HumanEval	Avg.
Vanilla	○	○	○	87.46	75.71	94.47	93.13	87.69
CoT (▷ <i>NeurIPS’22</i> [†])	○	○	○	88.74 _{↑1.28}	85.72 _{↑10.01}	94.85 _{↑0.38}	93.41 _{↑0.28}	90.68
ComplexCoT (▷ <i>ICLR’24</i> [†])	○	○	○	89.41 _{↑1.95}	86.45 _{↑10.74}	95.03 _{↑0.56}	93.14 _{↑0.02}	91.01
MoA (▷ <i>ICLR’25</i> [†])	●	○	○	89.77 _{↑2.31}	80.76 _{↑5.05}	94.77 _{↑0.30}	94.03 _{↑0.90}	89.83
GPTSwarm (▷ <i>ICML’24</i> [†])	●	●	◐	90.03 _{↑2.57}	90.12 _{↑14.41}	95.24 _{↑0.77}	94.16 _{↑1.03}	92.39
G-Designer (▷ <i>ICML’25</i> [†])	●	●	◐	90.85 _{↑3.39}	<u>90.54</u> _{↑14.83}	<u>96.66</u> _{↑2.19}	<u>94.82</u> _{↑1.69}	93.22
GoA (▷ <i>ICLR’26</i>)	●	●	◐	<u>91.53</u> _{↑4.07}	90.24 _{↑14.53}	95.43 _{↑0.96}	94.25 _{↑1.12}	92.86
SIGMA	●	●	●	92.16 _{↑4.70}	91.43 _{↑15.72}	96.81 _{↑2.34}	95.22 _{↑2.09}	93.91

tional modeling Rel., and partial conflict handling Conf., resulting in varying degrees of performance gains. For instance, G-Designer performs well on GSM8K and HumanEval but is still slightly behind GoA on MMLU. Finally, our method **SIGMA** achieves the best results across all four datasets, reaching 92.16% and 91.43% on MMLU and MMLU-Pro, respectively, with an improvement of 0.63 and 1.90 points over GoA, and demonstrating stable gains on GSM8K 96.81% and HumanEval 95.22%. These results strongly validate that our explicit trust–conflict modeling effectively enhances the reliability and consistency of multi-agent reasoning. Moreover, **SIGMA** simultaneously supports multi-agent interaction, inter-agent relational modeling and conflict-aware signal processing, highlighting its comprehensive adaptability and robustness in complex reasoning scenarios.

Experimental Analysis on GSM8K. Taking the GSM8K dataset as an example, the experiments employed multi-agent reasoning using DeepSeek-V3.2-based agents, with single-agent methods serving as a baseline for comparison. The dataset was divided into 659 batches, with an average batch processing time of approximately 73.79 seconds, and the overall average accuracy reached 96.81%. Outputs from each agent were integrated through **SIGMA**’s Signed Consensus, where positive and negative edges correspond to trust and conflict relationships, respectively, effectively mitigating the influence of low-confidence or conflicting information on the final answers.

For each problem, the agents’ reasoning outputs and corresponding weights were recorded and used to generate the final consensus answer. For instance, in the *Follower growth task*, the agents’ output weights were $[0.35, -0.20, 0.35, 0.10]$, and the weighted integration produced a total predicted follower count of 180, matching the ground truth; in the *French fries problem*, the final consensus output was 48, also consistent with the correct answer. Overall, **SIGMA** efficiently combines reasoning from multiple agents, suppresses conflicting signals, and substantially improves accuracy on the GSM8K dataset, demonstrating robust and consistent multi-agent collaborative reasoning performance.

Experimental Analysis on HumanEval. We evaluated **SIGMA** on the HumanEval dataset using DeepSeek-V3.2-based agents for multi-agent collaborative reasoning, with single-agent execution serving as a baseline. The dataset was divided into 80 batches, with an average batch processing time of approximately 65.4 seconds, and the final average functional correctness accuracy reached 95.22%. Outputs from each agent were integrated using **SIGMA**’s Signed Consensus, where positive and negative edges represent trust and conflict relationships, respectively, effectively mitigating the impact of low-confidence or conflicting outputs on the final answers.

For each problem, the reasoning outputs from all agents along with their corresponding weights were recorded and combined to generate the final consensus solution. For example, in a *Function-level task to determine whether a number is a simple power of another*, the agents’ output weights were $[0.45, -0.15, 0.30, 0.10, -0.10]$, and after weighted integration, the consensus output correctly returned True for `is_simple_power(8,2)` and False for `is_simple_power(3,2)`, matching the

expected results. In this process, Signed Consensus assigns higher weights to more reliable agents while suppressing low-confidence or conflicting outputs, effectively guiding the agents to reach the correct consensus. Similarly, in another string manipulation task, the final consensus output passed all internal test cases, confirming correct functionality. These results demonstrate that SIGMA not only integrates reasoning across multiple agents but also leverages the positive and negative edges in the signed graph to actively suppress erroneous signals, thereby improving the overall functional correctness of code generation. Overall, our approach exhibits robust and consistent multi-agent collaborative reasoning performance on the HumanEval dataset.

E.2 Hardware and Software Configurations.

We conduct the experiments using the following hardware and software configurations via AutoDL³ with SSH access through PyCharm Professional:

- **Operating System:** Ubuntu 20.04 LTS
- **CPU:** 14-core Intel(R) Xeon(R) Gold 6348 CPU @ 2.60GHz with 120GB RAM
- **GPU:** NVIDIA A800 80GB PCIe, CUDA 13.0, Driver Version 580.65.06
- **Storage:** System disk: 30GB; Data disk: 50GB SSD
- **Software:** Python 3.11, PyTorch 2.11.0, PyTorch Geometric 2.7.0, CUDA 13.0
- **Access:** Experiments run via SSH on AutoDL cloud instance using PyCharm Professional

F Additional Related Work

F.1 Signed Graph Representation Learning.

Signed graphs offer a principled framework to model interactions that can be either supportive or antagonistic, with edges encoding both polarity and magnitude [34, 32, 33]. Rooted in social balance theory [64, 65], early work emphasized structural consistency, such as identifying balanced and unbalanced triads, while more recent advances in graph neural networks extend message passing to signed edges [36, 66], enabling learned representations that account for agreement and conflict. These techniques have proven effective in social networks, recommendation systems, and opinion dynamics, capturing complex relational structures and mitigating propagation of misleading information. Despite these successes, signed graph methods remain largely unexplored in LLM-based multi-agent systems (MAS) for reasoning, where agents may produce heterogeneous and sometimes conflicting outputs. Leveraging signed graphs in MAS enables modeling of trust and conflict among agents, facilitating conflict-aware aggregation, robust multi-hop propagation, and improved consistency in multi-agent reasoning under heterogeneous interactions.

F.2 Conflict-Aware Multi-Agent Learning

In LLM-based multi-agent systems (MAS), handling conflicting or unreliable agents has received growing attention. Prior works investigate weighted aggregation [57, 59] and trust-aware message passing [25, 26] to mitigate the impact of adversarial or low-quality agents. These approaches improve robustness in specific tasks, such as collaborative decision-making or planning, by assigning higher influence to reliable agents while attenuating misleading signals from unreliable ones. However, they are often limited to fixed topologies or task-specific heuristics and rarely generalize to multi-hop, reasoning-intensive LLM-based scenarios. In contrast, our method leverages signed graph modeling to provide a principled and flexible framework that explicitly encodes both supportive and conflicting relationships, enabling conflict-aware aggregation and robust multi-agent reasoning under heterogeneous interactions.

³<https://autodl.com>

G Limitation

Focus on reasoning benchmarks with clear answers. **SIGMA** consistently outperforms existing methods across six benchmarks spanning general reasoning, mathematical reasoning, and code generation, demonstrating the core value of explicit trust–conflict modeling in tasks that require precise and verifiable reasoning. Currently, our evaluation focuses on these settings with well-defined correct outputs, while **SIGMA**’s performance on open-ended tasks such as creative writing, multi-turn dialogue, or strategic planning has not been specifically explored. In such tasks, the notion of "correctness" is inherently more ambiguous, and disagreements among agents can sometimes be a source of creativity rather than noise to be eliminated. Importantly, this scope does not diminish **SIGMA**’s advantages on mainstream reasoning tasks; rather, it opens up exciting opportunities for extending the signed graph approach to collaborative scenarios where open-ended, constructive conflict plays a central role.

H Broader Impacts

SIGMA significantly enhances the robustness and reliability of multi-agent reasoning by explicitly modeling trust, conflict, and neutral relations among agents through signed graphs. This capability offers direct benefits for high-stakes applications that demand precise reasoning, such as medical decision support, legal analysis, and scientific research, by effectively suppressing error propagation and dynamically isolating unreliable agents, thereby reducing the spread of misinformation in collaborative AI pipelines. Moreover, the framework’s interpretable trust dynamics strengthen the transparency and auditability of automated decision-making, providing meaningful support for the deployment of trustworthy AI.

On the other hand, any technology aimed at improving group decision quality carries potential for misuse. If improperly configured, **SIGMA**’s conflict-suppression mechanism may inadvertently silence constructive dissent or minority viewpoints in open-ended deliberation or opinion aggregation scenarios. In addition, robust multi-agent reasoning capabilities could be exploited to generate more persuasive disinformation. It is worth emphasizing that the signed graph formalism underlying **SIGMA** is inherently transparent: edge polarities and consensus weights can be inspected, adjusted, or overridden by human operators. We encourage future work to accompany deployment with appropriate human oversight and usage guidelines, so as to maximize the benefits of collaborative AI while minimizing potential risks.